

Khushal Mehta

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EDUCATION

University of Western Ontario
Bachelor of Science in Computer Science

Expected Graduation: Sep. 2026
London, Ontario

EXPERIENCE

Machine Learning Engineer Intern

May 2025 – April 2026

Zintlr AI

- Architected production-grade deep learning inference pipelines for cargo layout optimization, processing **10,000+ delivery orders** with sub-second latency using PyTorch and reinforcement learning algorithms
- Developed and deployed computer vision models on **AWS SageMaker** for intelligent dispatch optimization, achieving **25% reduction** in route planning computational overhead across large-scale logistics networks
- Implemented MLOps practices with automated model monitoring and CI/CD pipelines on **AWS Lambda/EC2**, ensuring seamless production deployment workflows

PROJECTS

Conscient | *uOttawa Hacks 2025* | **React, MongoDB, Meta's Llama 3, Random Forest** **January 2025**

- Developed AI-powered mental health platform combining **React.js** frontend with **MongoDB Atlas** for secure user authentication and scalable data storage architecture
- Achieved **97% accuracy** in mood analysis by training a **Random Forest classifier** on text embeddings generated by **Meta's Llama 3 model**, significantly outperforming baseline accuracy metrics by **12 percentage points**
- Integrated self-learning AI chatbot with smart diary system using **Groq API**, enhancing conversational context and enabling peer connection through AI-determined compatibility matching

PennyWise | Live Demo | **Next.js, TypeScript, Google Gemini API, Tailwind CSS** **June 2025**

- Developed intelligent personal finance platform with **AI-powered spending analysis** and behavioral profiling, providing actionable financial recommendations through **Google Gemini API** integration
- Built comprehensive expense tracking system with **recurring expense detection** and smart budget alerts, enabling real-time financial monitoring and **proactive spending warnings**
- Implemented **Next.js + TypeScript** architecture with trend analysis algorithms, delivering personalized financial insights and spending personality profiles for enhanced user financial awareness

Simulated Souls | **React, TypeScript, Google Gemini, Cloudflare AI** **July 2025**

- Engineered an immersive AI-powered text adventure game integrating **Google's Gemini** for dynamic storytelling and **Cloudflare's stable-diffusion-xl-lightning** for real-time image generation
- Architected responsive **React 19 + TypeScript** frontend with **Express.js** proxy server, achieving seamless story progression and **sub-second** AI response times
- Implemented advanced **CORS handling** and optimized AI service integration, delivering interactive gameplay with contextual narratives and accompanying artwork generation

CERTIFICATIONS

Artificial Intelligence Design and Engineering in Vertex AI by Google Cloud | *AI Design* **April 2024**

- Mastered **generative AI** and **ML workflows** on Vertex AI.
- Skilled in real-world AI solutions, **predictive modeling**, and **prompt engineering**.

Attention Mechanism in Machine Learning by Google Cloud | *Machine Learning* **April 2024**

- Learned **attention mechanisms** for **AI** and **ML applications**.
- Proficient in designing AI systems with advanced attention models.

SKILLS & TECHNOLOGIES

Languages: Python, JavaScript/TypeScript, Java, C++

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

Backend: Node.js, Express.js, Flask, Socket.io

Machine Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Random Forest, Attention Mechanisms

AI APIs & Models: Meta Llama 3, Google Gemini, Groq API, Cloudflare AI

Cloud & Databases: AWS (SageMaker, Lambda, EC2), MongoDB Atlas